The documentation and process conversion measures necessary to comply with this revision shall be completed by:
28 December 2018.

INCH-POUND

MIL-PRF-19500/456F
28 August 2018
SUPERSEDING
MIL-PRF-19500/456E
6 March 2009

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, NPN, SILICON, HIGH-POWER,
TYPES 2N5302 AND 2N5303, CASE MOUNT THROUGH HOLE PACKAGE,
QUALITY LEVELS JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon, high-power transistors. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each device type as specified in MIL-PRF-19500.

1.2 Package outline. The device package for the encapsulated device types is a TO-204AA (similar to TO-3) in accordance with figure 1.

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

1.3.1 Ratings applicable to all Part or Identifying Numbers (PIN). T_J = T_{STG} = –65°C to +200°C.

1.3.2 Ratings applicable to individual types.

Type	P_T (1) T_A = +25°C	P_T (1) T_C = +100°C	V_{CBO}	V_{CEO}	V_{EBO}	I_B	I_C	$R_{\theta JC}$
	W	W	V dc	V dc	V dc	A dc	A dc	°C/W Max
2N5302	5	115	60	60	5.0	7.5	30	0.875
2N5303	5	115	80	80	5.0	7.5	20	0.875

(1) Derate linearly, 1.14 mW/°C above T_C = +100°C.
Derate linearly, 28.57 mW/°C above T_A = +25°C.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.4 Primary electrical characteristics. Unless otherwise specified, T_A = +25°C.

	h_{FE2} (1) V_{CE} = 2 V dc I_C = 15 A dc	h_{FE2} (1) V_{CE} = 2 V dc I_C = 10 A dc	$\lvert h_{fe} \rvert$ V_{CE} = 10 V dc I_C = 1 A dc f = 1 MHz	$V_{BE(sat)2}$ (1) I_C = 15 A dc I_B = 1.5 A dc		$V_{CE(sat)2}$ (1) I_C = 15 A dc I_B = 1.5 A dc		C_{obo} V_{CB} = 10 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz	Pulse response	
									t_{on}	t_{off}
	2N5302	2N5303		2N5302	2N5303	2N5302	2N5303			
			V dc	V dc	V dc	V dc	V dc	pF	µs	µs
Min	15	15	2							
Max	60	60	40	1.8	2	1	1.5	800	1.1	3.0

(1) Pulsed (see 4.5.1).

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", "JANTXV", and "JANS".

1.5.2 Device type. The designation system for the devices covered by this specification sheet is as follows.

1.5.2.1 First number and first letter symbols. The devices of this specification sheet use the first number and letter symbols "2N".

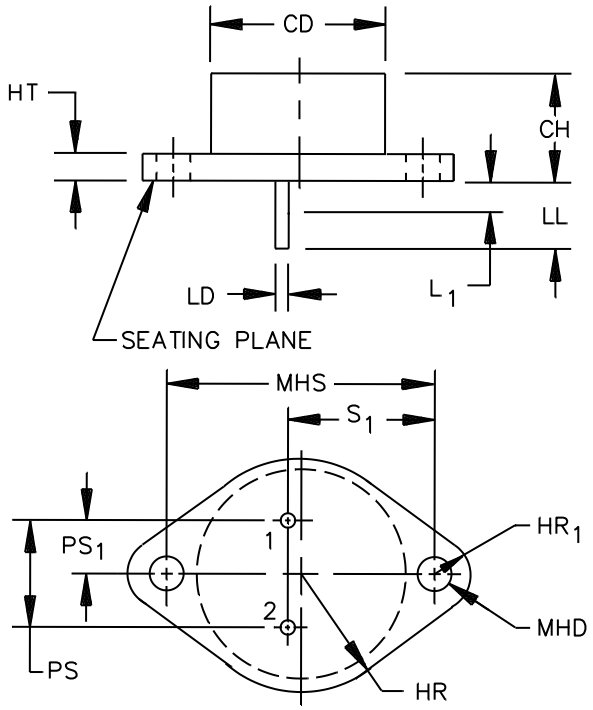
1.5.2.2 Second number symbols. The second number symbols for the devices covered by this specification sheet are as follows: "5302" and "5303".

1.5.3 Suffix symbols. Suffix symbols are not applicable to this specification sheet.

1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD		.875		22.23	5
CH	.270	.380	6.86	8.89	
HR	.495	.525	12.57	13.34	6
HR_1	.131	.188	3.33	4.78	6
HT	.060	.135	1.52	3.43	
L_1		.050		1.27	7
LD	.038	.043	0.97	1.09	7, 8, 9
LL	.312	.500	7.92	12.70	7, 8, 9
MHD	.151	.161	3.84	4.09	10
MHS	1.177	1.197	29.90	30.40	
PS	.420	.440	10.67	11.18	9
PS_1	.205	.225	5.21	5.72	7, 9
S_1	.655	.675	16.64	17.15	9



NOTES:

1. Dimensions are in inches. Millimeters are given for general information only.
2. Terminal 1 is the emitter and terminal 2 is the base. The collector shall be electrically connected to the case.
3. Mounting holes shall be deburred on the seating plane side.
4. The seating plane of the header shall be flat within .001 inch (0.03 mm) concave to .004 inch (0.10 mm) convex inside a .930 inch (23.62 mm) diameter circle on the center of the header and flat within .001 inch (0.03 mm) concave to .006 inch (0.15 mm) convex overall.
5. Body contour is optional within zone defined by dimension CD.
6. Applies to all sides (dimension HR) or both ends (dimension HR_1).
7. Applies to both terminals.
8. Dimension LD applies between dimension L_1 and dimension LL. Lead diameter shall not exceed twice dimension LD within dimension L_1.
9. Measurement for this dimension shall be taken at points .050 inch (1.27 mm) to .055 inch (1.40 mm) below the seating plane. When gauge is not used, measurement shall be made at seating plane.
10. Applies to both holes.
11. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions for TO-204AA (similar to TO-3).

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 – Test Methods For Semiconductor Devices.

(Copies of these documents are available online at http://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Pin-out. The pin-out of the device package shall be as shown on figure 1. Terminal 1 is the emitter, terminal 2 is the base, and the collector shall be electrically connected to the case.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

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3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.3 Screening (quality levels JANS, JANTX, and JANTXV only). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurement	
	JANS level	JANTX and JANTXV levels
3c (1)	Thermal impedance (see 4.3.2)	Thermal impedance (see 4.3.2)
9	I_{CEX1} and h_{FE2}	Not applicable
11	I_{CEX1} and h_{FE2} ΔI_{CEX1} = 100 percent or 1 µA whichever is greater; Δh_{FE2} = ±20 percent	I_{CEX1} and h_{FE2}
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; ΔI_{CEX1} = +100 percent of initial value or 1 µA, whichever is greater. Δh_{FE2} = ±15 percent of initial value.	Subgroup 2 of table I herein; ΔI_{CEX1} = +100 percent of initial value or 1 µA, whichever is greater. Δh_{FE2} = ±15 percent of initial value.

(1) Thermal impedance shall be performed anytime after sealing provided temperature cycling is performed in accordance with MIL-PRF-19500, screen 3a; JANTX and JANTXV levels do not need to be repeated in screening requirements.

4.3.1 Power burn-in conditions. The power burn-in conditions shall be as follows: T_J = +187.5°C ±12.5°C, $V_{CE} \geq$ 10 V dc, $T_A \leq$ +35°C.

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4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{MD}, t_{SW} (V_C and V_H where appropriate). See table II, subgroup 4 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E–V of MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in tables E-VIA (JANS) and E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and 4.4.2.1 and 4.4.2.2 herein.

4.4.2.1 Quality level JANS.

Subgroup	Method	Condition
B4	1037	V_{CB} = 20 V dc; P_D = 5 W at T_A = room ambient as defined in 4.5 of MIL-STD-750; $t_{on} = t_{off}$ = 3 minutes minimum for 2,000 cycles. No heat sink or forced air cooling on devices shall be permitted.
B5	1027	V_{CB} = 20 V dc; T_A = +125°C ±25°C for 96 hours; P_T = 5 W at T_A = +125°C or adjusted as required by the chosen T_A to give an average lot, T_J = +275°C.

4.4.2.2 Quality levels JAN, JANTX, and JANTXV.

Subgroup	Method	Condition
B3	1037	For solder die attach: $V_{CB} \geq$ 10 V dc; $T_A \leq$ 35°C, 2,000 cycles. No heat sink or forced air cooling on devices shall be permitted.
B3	1026	For eutectic die attach: $V_{CB} \geq$ 10 V dc; $T_A \leq$ 35°C, adjust P_T to achieve T_J = 175°C minimum.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and as follows.

Subgroup	Method	Condition
C6	1037	For solder die attach: $V_{CB} \geq$ 10 V dc; $T_A \leq$ 35°C, 6,000 cycles. No heat sink or forced air cooling on devices shall be permitted.
C6	1026	For eutectic die attach: $V_{CB} \geq$ 10 V dc; $T_A \leq$ 35°C, adjust P_T to achieve T_J = 175°C minimum.

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4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with method 3131 of MIL-STD-750. The following details shall apply:

a. Collector current magnitude during power application shall be 3.14 A dc.

b. Collector to emitter voltage magnitude shall be 20 V dc.

c. Reference temperature measuring point shall be the case.

d. Reference point temperature shall be $+25°C \leq T_R \leq +75°C$ and recorded before the test is started.

e. Mounting arrangement shall be with heat sink to header.

f. Maximum limit of $R_{\theta JC}$ shall be 0.875°C/W.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance 2/	3131	See 4.3.2	$Z_{\theta JX}$			°C/W
Collector - emitter breakdown voltage	3011	Bias condition D; I_C = 200 mA dc pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N5302				60		V dc
2N5303				80		V dc
Collector - emitter cutoff current	3041	Bias condition D;	I_{CEO}		10.0	µA dc
2N5302		V_{CE} = 60 V dc				
2N5303		V_{CE} = 80 V dc				
Emitter - base cutoff current	3061	Bias condition D; V_{EB} = 5 V dc	I_{EBO}		5.0	µA dc
Collector - emitter cutoff current	3041	Bias condition A; V_{BE} = –1.5 V dc	I_{CEX1}		5.0	µA dc
2N5302		V_{CE} = 60 V dc				
2N5303		V_{CE} = 80 V dc				
Collector - base cutoff current	3036	Bias condition D;	I_{CBO}		5.0	µA dc
2N5302		V_{CE} = 60 V dc				
2N5303		V_{CE} = 80 V dc				
Base - emitter saturated voltage	3066	Test condition A; I_C = 10 A dc; I_B = 1 A dc; pulsed (see 4.5.1)	$V_{BE(sat)1}$		1.7	V dc
Base - emitter saturated voltage	3066	Test condition A; I_C = 15 A dc; I_B = 1.5 A dc; pulsed (see 4.5.1)	$V_{BE(sat)2}$			
2N5302					1.8	V dc
2N5303					2.0	V dc
Base - emitter saturated voltage	3066	Test condition A; I_C = 20 A dc; pulsed (see 4.5.1);	$V_{BE(sat)3}$			
2N5302		I_B = 2 A dc			2.5	V dc
2N5303		I_B = 4 A dc			2.5	V dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Base - emitter voltage (nonsaturated)	3066	Test condition B; V_{CE} = 2 V dc; pulsed (see 4.5.1);	$V_{BE(on)1}$			
2N5302		I_C = 15 A dc			1.8	V dc
2N5303		I_C = 10 A dc			1.5	V dc
Base - emitter voltage (unsaturated)	3066	Test condition B; V_{CE} = 4 V dc, pulsed (4.5.1);	$V_{BE(on)2}$			
2N5302		I_C = 30 A dc			3.0	V dc
2N5303		I_C = 20 A dc			2.5	V dc
Collector - emitter saturated voltage	3071	I_C = 10 A dc; I_B = 1 A dc; pulsed (see 4.5.1)	$V_{CE(sat)1}$			
2N5302					0.75	V dc
2N5303					1.0	V dc
Collector - emitter saturated voltage	3071	I_C = 15 A dc; I_B = 1.5 A dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$			
2N5302					1.0	V dc
2N5303					1.5	V dc
Collector - emitter saturated voltage	3071	I_C = 20 A dc; pulsed (see 4.5.1);	$V_{CE(sat)3}$			
2N5302		I_B = 2 A dc			2.0	V dc
2N5303		I_B = 4 A dc			2.0	V dc
Collector - emitter saturated voltage	3071	I_C = 30 A dc; pulsed (see 4.5.1);	$V_{CE(sat)4}$			
2N5302 (only)		I_B = 6 A dc			3.0	V dc
Forward-current transfer ratio	3076	V_{CE} = 2 V dc; I_C = 1 A dc; pulsed (see 4.5.1)	h_{FE1}	40		
Forward-current transfer ratio	3076	V_{CE} = 2 V dc; pulsed (see 4.5.1);	h_{FE2}			
2N5302		I_C = 15 A dc		15	60	
2N5303		I_C = 10 A dc		15	60	
Forward-current transfer ratio	3076	V_{CE} = 4 V dc; pulsed (see 4.5.1);	h_{FE3}			
2N5302		I_C = 30 A dc		5		
2N5303		I_C = 20 A dc		5		

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High-temperature operation:		T_A = +150°C				
Collector to emitter cutoff current 2N5302 2N5303	3041	Bias condition A; V_{BE} = –1.5 V dc V_{CE} = 60 V dc V_{CE} = 80 V dc	I_{CEX2}		50	µA dc
Low-temperature operation:		T_A = -55°C				
Forward current transfer ratio 2N5302 2N5303	3076	V_{CE} = 2 V dc; pulsed (see 4.5.1); I_C = 15 A dc I_C = 10 A dc	h_{FE4}	 7 7		
Subgroup 4						
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = 10 V dc; I_C = 1 A dc; f = 1 kHz	h_{fe}	40	240	
Magnitude of small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 1 A dc; f = 1 MHz	$\lvert h_{fe} \rvert$	2	40	
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz < f ≤ 1 MHz	C_{obo}		800	pF
Switching parameters:						
Pulse delay time		See figure 2	t_d		0.2	µs
Pulse rise time		See figure 2	t_r		0.9	µs
Pulse storage time		See figure 2	t_s		2.0	µs
Pulse fall time		See figure 2	t_f		1.0	µs
Subgroup 5						
Safe operating area (continuous dc)	3051	T_C = 25°C; power application time ≥ 1 sec, 1 cycle (see figures 3 and 4)				
Test 1						
2N5302 2N5303		V_{CE} = 6.67 V dc; I_C = 30 A dc V_{CE} = 10 V dc; I_C = 20 A dc				

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5 - Continued						
Test 2						
2N5302, 2N5303		V_{CE} = 20 V dc; I_C = 10 A dc				
Test 3						
2N5302, 2N5303		V_{CE} = 40 V dc; I_C = 3 A dc				
Test 4						
2N5302 2N5303		V_{CE} = 50 V dc; I_C = 600 mA dc V_{CE} = 60 V dc; I_C = 600 mA dc				
Safe operating area (clamped switching) 2N5302 2N5303	3053	Condition B, T_A = 25°C; V_{CE} = 15 V dc; (see figures 5 and 6) Clamp voltage = 60 V dc; I_C = 30 A dc Clamp voltage = 80 V dc; I_C = 20 A dc				
Electrical measurements		See table I, subgroup 2				
Subgroups 6 and 7						
Not applicable						

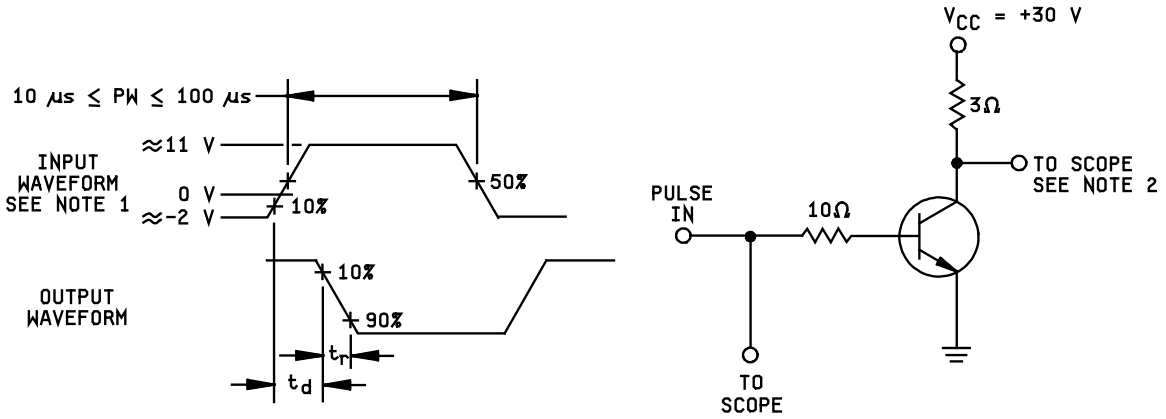
1/ For sampling plan see MIL-PRF-19500.
2/ For end-point electrical measurements, this test is required for the following subgroups:
Group B, subgroups 3, 4, and 5 (JANS).
Group B, subgroups 2 and 3 (JAN, JANTX, AND JANTXV).
Group C, subgroups 2 and 6.
Group E, subgroup 1.

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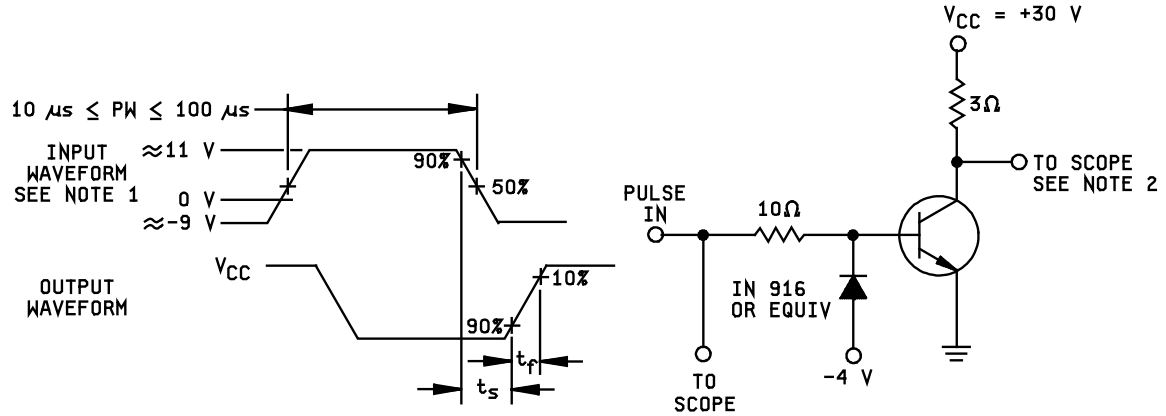
TABLE II. Group E inspection (all quality levels) - for qualification or re-qualification only.

Inspection	MIL-STD-750		Sample plan
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling	1051	Test condition C, 500 cycles.	
Hermetic seal Fine leak Gross leak	1071		
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 2			45 devices c = 0
Blocking life	1048	Test temperature = +125°C; V_{CB} = 80 percent of rated, without going over the maximum rated V_{CEO}; t = 1,000 hours.	
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 4			
Thermal impedance curves		See MIL-PRF-19500.	Sample size N/A
Subgroup 6			
ESD	1020		
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition B.	

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t_d AND t_r TIME TEST CIRCUIT



t_s AND t_f TIME TEST CIRCUIT

NOTES:

1. The rise time (t_r) of the applied pulse shall be ≤ 2 ns; duty cycle ≤ 2 percent; generator source impedance shall be 50 Ω.
2. Output sampling oscilloscope: Z_{IN} ≥ 100 kΩ; C_{IN} ≤ 12 pF; rise time ≤ 0.2 ns.

FIGURE 2. Pulse response test circuit.

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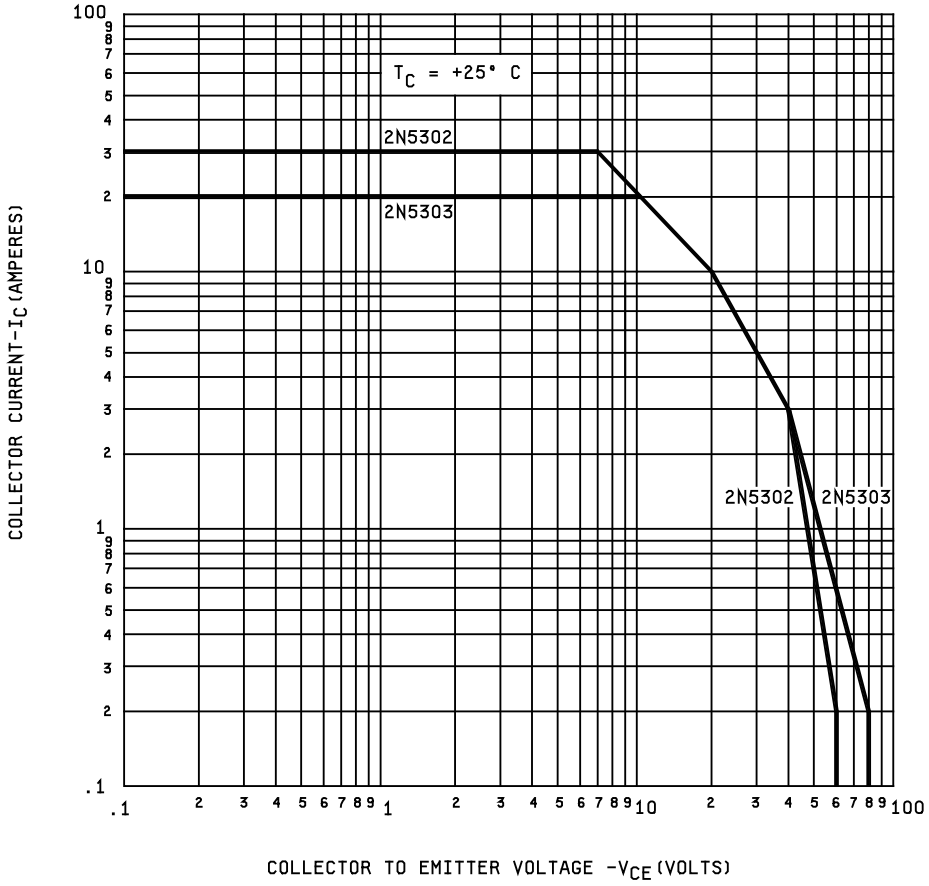


FIGURE 3. Maximum safe operating area graph dc.

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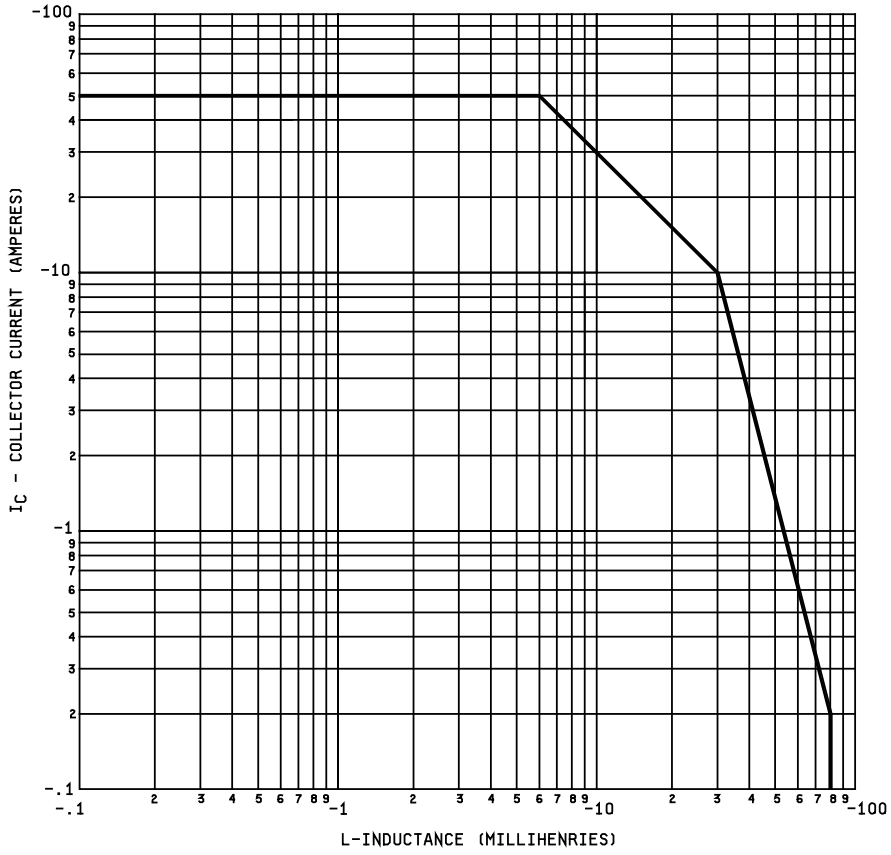
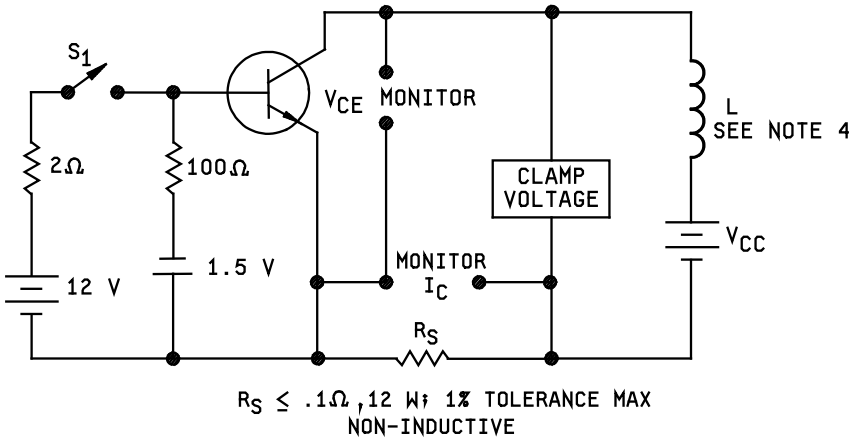


FIGURE 4. Safe operating area for switching between saturation and cutoff (unclamped inductive load).

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L = 2.0 mH (2 each 1 mH, 50 A, .001 Ω, Sanford Miller CK-50, or equivalent).

Procedure:

1. With switch S_1 closed, set the specified test conditions.
2. Open S_1. Device fails if the clamp voltage if not reached.
3. Perform specified endpoints tests.

FIGURE 5. Clamped inductive sweep test circuit.

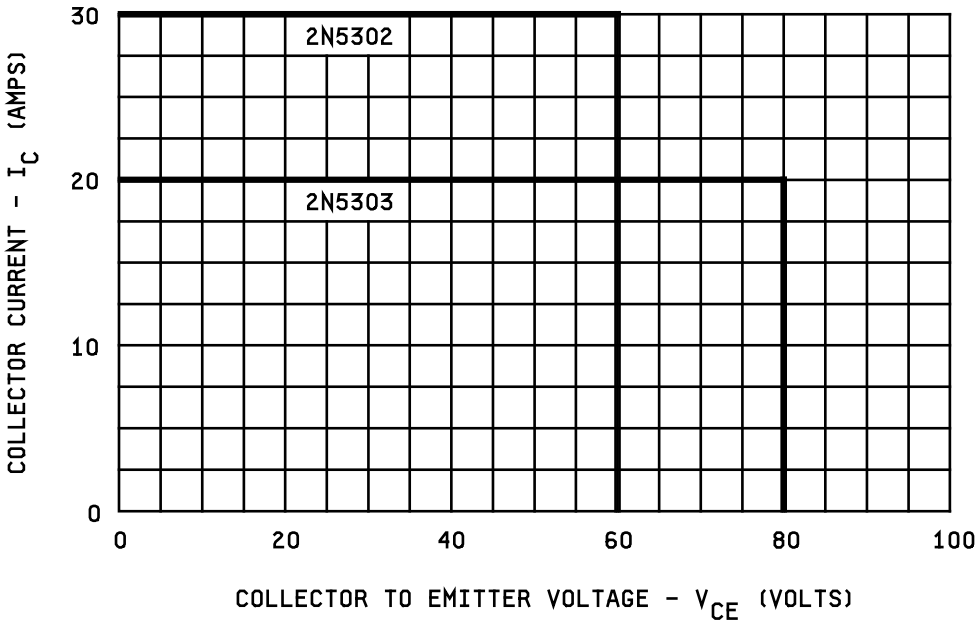


FIGURE 6. Safe operating area for switching between saturation and cutoff (clamped inductive load).

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

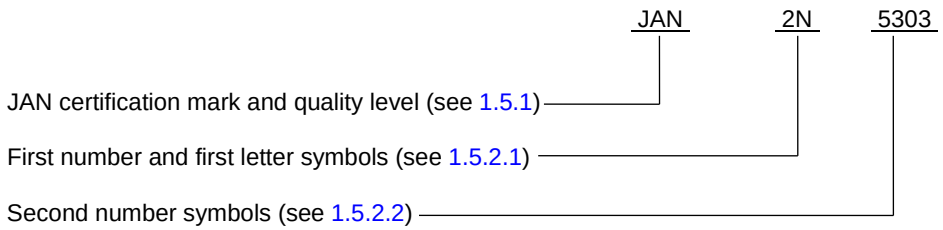
b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.5.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at http://qpldocs.dla.mil.

6.4 PIN construction example. The PINs for encapsulated devices are constructed using the following form.



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6.5 List of PINs. The following is a list of possible PINs available for encapsulated devices covered by this specification sheet.

JAN quality level	JANTX quality level	JANTXV quality level	JANS quality level
JAN2N5302	JANTX2N5302	JANTXV2N5302	JANS2N5302
JAN2N5303	JANTX2N5303	JANTXV2N5303	JANS2N5303

6.6 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at semiconductor@dla.mil or by facsimile (614) 693-1642 or DSN 850-6939.

6.7 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - EC
Air Force - 19, 71

Preparing activity:
DLA - CC

(Project 5961-2018-066)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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